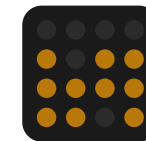


How to use these token cutouts



These pages contain the text *The Cat in the Hat* by Dr Seuss. Each **cutout** shows a **next word** paired with the **previous 2 words** that came immediately before it in the original text.

Model vitals			
1962	0.84		
tokens	bits/token entropy		
1.8	1.47		
perplexity	branching factor		

Anatomy of a cutout



Every distinct word has its own colour. **Previous** words appear inside a coloured box (the word’s own colour as the background, with white text); the free-standing **next word** appears in plain coloured text. The same word always wears the same colour, whether you see it inside a box or free-standing. Two unrelated words can occasionally share a colour, so always verify the word itself matches—not just the colour.

Setup

Cut out the tokens along the dotted lines and **spread them out** face-up on the table, with no overlap if possible. This is the “training” step: every (previous, next) combination from the original text is now a physical cutout sitting on your table.

How to play: the matching game

To generate text, repeatedly find a cutout whose **previous 2 words** matches the last 2 words you’ve written, then **write its next word** onto your page. The word you just wrote becomes part of what you’ll match against next—that’s the chain.

Worked example

Step 1. Pick any cutout to begin—say this one—and copy its previous 2 words *and* its next word onto your page.

Step 2. Look at the last 2 words you’ve written (*sun did*) and find a cutout whose previous 2 words match them. Write its next word.

Step 3. Repeat: find a cutout whose previous 2 words match the last 2 words you’ve written, and add its next word.

Keep going—write as much or as little as you like.

the cutout

the sun did

sun did not

did not shine

your page

the sun did

the sun did not

the sun did not shine

Tips

Use colour as a fast filter. Scan by the rightmost previous-word box’s colour first, then verify the actual word matches.

Pick from many candidates by eye. The more cutouts share the same previous 2 words, the more often your eye lands on one—that’s **weighted sampling** for free, and it’s why some words follow others more often in your text.

Put the cutout back after using it—removing it would change the model’s distribution next time round.

the sun **the sun** did **sun** **did** not **did** **not** shine

not **shine** . **shine** . it . **it** was **it** was too

was **too** wet **too** **wet** to **wet** **to** play **to** **play** .

play . so . **so** we **so** **we** sat **we** **sat** in **sat** **in** the

in **the** house **the** **house** all **house** **all** that

all **that** cold **that** **cold** , **cold** , cold , **cold** ,

cold , wet , **wet** day **wet** **day** . **day** . I . **I** sat

I sat there **sat** **there** with **there** **with** Sally

with **Sally** . **Sally** . we . **we** sat **we** **sat** there

sat **there** , **there** , we , **we** two **we** **two** .

two . and . and I and I said I said , said , how
, how I how I wish I wish we wish we had
we had something had something to
something to do to do ! do ! too ! too wet
too wet to wet to go to go out go out and
out and too and too cold too cold to
cold to play to play ball play ball . ball . so
. so we so we sat we sat in sat in the
in the house the house . house . we . we did
we did nothing did nothing at nothing at all

at all . all . so . so all so all we all we could

we could do could do was do was to was to sit

to sit ! sit ! sit ! sit ! sit ! sit ! sit ! sit

! sit ! sit ! and ! and we and we did we did not

did not like not like it like it . it . not

. not one not one little one little bit little bit .

bit . and . and then and then something

then something went something went bump

went bump ! bump ! how ! how that

how that bump that bump made

bump made us made us jump us jump !

jump ! we ! we looked we looked !

looked ! then ! then we then we saw

we saw him saw him step him step in

step in on in on the on the mat the mat !

mat ! we ! we looked we looked ! looked ! and

! and we and we saw we saw him saw him !

him ! the ! the cat the cat in cat in the

in the hat the hat ! hat ! and ! and he

and he said he said to said to us to us ,

us, Why, Why do Why do you do you sit

you sit there sit there like there like that

like that ? that ? I ? I know I know it

know it is it is wet is wet and wet and the

and the sun the sun is sun is not is not sunny

not sunny . sunny . But . But we But we can

we can have can have Lots have Lots of

Lots of good of good fun good fun that

fun that is that is funny is funny ! funny ! I

! I know I know some know some good

some good games good games we

games we could we could play could play ,

play , said , said the said the cat the cat .

cat . I . I know I know some know some new

some new tricks new tricks , tricks , said

, said the said the cat the cat in cat in the

in the hat the hat . hat . a . a lot a lot of

lot of good of good tricks good tricks .

tricks . I . I will I will show will show them

show them to them to you to you . you . your

. your mother your mother will

mother will not will not mind not mind at

mind at all at all if all if I if I do I do .

do . then . then Sally then Sally and

Sally and I and I did I did not did not know

not know what know what to what to say

to say . say . our . our mother our mother was

mother was out was out of out of the

of the house the house For house For the

For the day the day . day . But . But our

But our fish our fish said fish said , said , no

, no ! no ! no ! no ! no ! Make ! Make that

Make that cat that cat go cat go away

go away ! away ! tell ! tell that tell that cat

that cat in cat in the in the hat the hat you

hat you do you do not do not want

not want to want to play to play . play . he

. he should he should not should not be

not be here be here . here . he . he should

he should not should not be not be about

be about . about . he . he should he should not

should not be not be here be here When

here When your When your mother

your mother is mother is out is out !

out ! now ! now ! now ! now !

now ! have ! have no have no fear no fear .

fear . have . have no have no fear no fear !

fear ! said ! said the said the cat the cat .

cat . my . my tricks my tricks are

tricks are not are not bad not bad , bad , said

, said the said the cat the cat in cat in the
in the hat the hat . hat . Why . Why ,
Why , we , we can we can have can have Lots
have Lots of Lots of good of good fun
good fun , fun , if , if you if you wish
you wish , wish , with , with a with a game
a game that game that I that I call I call up
call up up up up up with up with a
with a fish a fish ! fish ! put ! put me
put me down me down ! down ! said ! said the

said the fish the fish . fish . this . this is

this is no is no fun no fun at fun at all

at all ! all ! put ! put me put me down

me down ! down ! said ! said the said the fish

the fish . fish . I . I do I do not do not wish

not wish to wish to fall to fall ! fall ! have

! have no have no fear no fear ! fear ! said

! said the said the cat the cat . cat . I . I will

I will not will not let not let you let you fall

you fall . fall . I . I will I will hold

will hold you hold you up you up high

up high as high as I as I stand I stand on

stand on a on a ball a ball . ball . with

. with a with a book a book one book one one

one one hand one hand ! hand ! and ! and a

and a cup a cup on cup on my on my hat

my hat ! hat ! But ! But that But that is

that is not is not all not all I all I can I can do

can do ! do ! said ! said the said the cat

the cat . cat look . look at

look at me at me ! me ! look ! look at

look at me at me now me now ! now ! said

! said the said the cat the cat . cat . with

. with a with a cup a cup and cup and a

and a cake a cake on cake on the on the top

the top of top of my of my hat my hat !

hat ! I ! I can I can hold can hold up

hold up two up two books two books !

books ! I ! I can I can hold can hold up

hold up the up the fish the fish ! fish ! and

! and a and a little a little toy little toy ship

toy ship ! ship ! and ! and some and some milk

some milk on milk on a on a dish a dish !

dish ! and ! and look and look ! look ! I

! I can I can hop can hop up hop up and

up and down and down on down on the

on the ball the ball ! ball ! But ! But that

But that is that is not is not all not all !

all ! Oh ! Oh , Oh , no , no . no . that . that is

that is not is not all not all . all look

. look at look at me at me ! me ! look

! look at look at me at me ! me ! look

! look at look at me at me now me now !

now ! it ! it is it is fun is fun to fun to have

to have fun have fun But fun But you

But you have you have to have to know

to know how know how . how . I . I can

I can hold can hold up hold up the up the cup

the cup and cup and the and the milk

the milk and milk and the and the cake

the cake ! cake ! I ! I can I can hold

can hold up hold up these up these books

these books ! books ! and ! and the

and the fish the fish on fish on a on a rake

a rake ! rake ! I ! I can I can hold

can hold the hold the toy the toy ship

toy ship and ship and a and a little a little toy

little toy man toy man ! man ! and ! and look

and look ! look ! with ! with my with my tail

my tail I tail I can I can hold can hold a

hold a red a red fan red fan ! fan ! I ! I can

I can fan can fan with fan with the

with the fan the fan as fan as I as I hop

I hop on hop on the on the ball the ball !

ball ! But ! But that But that is that is not

is not all not all . all . Oh . Oh , Oh , no

, no . no . that . that is that is not is not all

not all . all that . that is that is what

is what the what the cat the cat said cat said .

said then . then he then he fell

he fell on fell on his on his head his head !

head ! he ! he came he came down

came down with down with a with a bump

a bump From bump From up From up there

up there on there on the on the ball the ball .

ball . and . and Sally and Sally and I

and I , I , we , we saw we saw all saw all the

all the things the things fall things fall !

fall ! and ! and our and our fish our fish came

fish came down came down , down , too

, too . too . he . he fell he fell into fell into a

into a pot a pot ! pot ! he ! he said he said ,

said , do , do I do I like I like this like this ?

this ? Oh ? Oh , Oh , no , no ! no ! I ! I do

I do not do not . not . this . this is this is not

is not a not a good a good game good game ,

game , said , said our said our fish our fish as

fish as he as he lit he lit . lit . no . no , no , I

, I do I do not do not like not like it like it ,

it , not , not one not one little one little bit

little bit ! bit ! now ! now look now look what

look what you what you did you did !

did ! said ! said the said the fish the fish to

fish to the to the cat the cat . cat . now

. now look now look at look at this

at this house this house ! house ! look

! look at look at this at this ! this ! look

! look at look at that at that ! that ! you

! you sank you sank our sank our toy

our toy ship toy ship , ship , sank , sank it

sank it deep it deep in deep in the in the cake

the cake . cake . you . you shook

you shook up shook up our up our house

our house and house and you and you bent

you bent our bent our new our new rake

new rake . rake . you . you should

you should not should not be not be here

be here When here When our

When our mother our mother is mother is not

is not . not . you . you get you get out

get out of out of this of this house

this house ! house ! said ! said the

said the fish the fish in fish in the in the pot

the pot . pot . But . But I But I like I like to

like to be to be here be here . here . Oh . Oh ,

Oh , I , I like I like it like it a it a lot a lot !

lot ! said ! said the said the cat the cat in

cat in the in the hat the hat to hat to the

to the fish the fish in fish in the in the pot

the pot . pot . I . I will I will not will not go

not go away go away . away . I . I do I do not

do not wish not wish to wish to go to go !

go ! and ! and so and so , so , said , said the

said the cat the cat in cat in the in the hat

the hat , hat , so , so so so so so . so .

. . . . I . I will I will show will show you

show you another you another good

another good game good game that

game that I that I know I know ! know ! and

! and then and then he then he ran he ran out

ran out . out . and . and , and , then , then ,

then , fast , fast as fast as a as a fox a fox ,

fox , the , the cat the cat in cat in the

in the hat the hat came hat came back

came back in back in with in with a

with a box a box . box . a . a big a big red

big red wood red wood box wood box . box . it

. it was it was shut was shut with shut with a

with a hook a hook . hook . now . now look

now look at look at this at this trick

this trick , trick , said , said the said the cat

the cat . cat . take . take a take a look a look !

look ! then ! then he then he got he got up

got up on up on top on top with top with a

with a tip a tip of tip of his of his hat

his hat . hat . I . I call I call this

call this game this game fun game fun in

fun in a in a box a box , box , said , said the

said the cat the cat . cat . in . in this

in this box this box are box are two

are two things two things I things I will

I will show will show to show to you

to you now you now . now . you . you will

you will like will like these like these two

these two things two things , things , said

, said the said the cat the cat with cat with a

with a bow a bow . bow . I . I will I will pick

will pick up pick up the up the hook

the hook . hook . you . you will you will see

will see something see something new

something new . new . two . two things

two things . things . and . and I and I call

I call them call them Thing them Thing one

Thing one and one and Thing and Thing two

Thing two . two . these . these things

these things will things will not will not bite

not bite you bite you . you . they . they want

they want to want to have to have fun

have fun . fun . then . then , then , out

, out of out of the of the box the box came

box came Thing came Thing two

Thing two and two and Thing and Thing one

Thing one ! one ! and ! and they and they ran

they ran to ran to us to us fast us fast .

fast . they . they said they said , said , how

, how do how do you do you do you do ?

do ? would ? would you would you like

you like to like to shake to shake hands

shake hands with hands with Thing

with Thing one Thing one and one and Thing

and Thing two Thing two ? two ? and

? and Sally and Sally and Sally and I and I did

I did not did not know not know what

know what to what to do to do . do . so

. so we so we had we had to had to shake

to shake hands shake hands with

hands with Thing with Thing one

Thing one and one and Thing and Thing two

Thing two . two . we . we shook

we shook their shook their two

their **two** hands **two** **hands** . **hands** . But

. But our But **our** **fish** **our** **fish** said **fish** **said** ,

said , **no** , **no** ! **no** ! **no** ! **no** ! those

! **those** things **those** **things** should

things **should** not **should** **not** **be** **not** **be** in

be **in** this **in** **this** house **this** **house** !

house ! Make ! **Make** them **Make** **them** go

them **go** ! **go** ! **they** ! **they** should

they **should** not **should** **not** **be** **not** **be** here

be **here** When **here** **When** your

When your mother your mother is

mother is not is not ! not ! put ! put them

put them out them out ! out ! put ! put them

put them out them out ! out ! said ! said the

said the fish the fish in fish in the in the pot

the pot . pot . have . have no have no fear

no fear , fear , little , little fish little fish ,

fish , said , said the said the cat the cat in

cat in the in the hat the hat . hat . these

. these things these things are things are good

are good things good things . things . and

. and he and he gave he gave them

gave them a them a pat a pat . pat . they

. they are they are tame are tame . tame . Oh

. Oh , Oh , so , so tame so tame ! tame ! they

! they have they have come have come here

come here to here to play to play . play . they

. they will they will give will give you

give you some you some fun some fun on

fun on this on this wet this wet , wet , wet

, wet , wet , wet , wet day wet day . day . now

. now , now , here , here is here is a is a game

a game that game that they that they like

they like , like , said , said the said the cat

the cat . cat . they . they like they like to

like to fly to fly kites fly kites , kites , said

, said the said the cat the cat in cat in the

in the hat the hat . hat . no . no ! no ! not

! not in not in the in the house the house !

house ! said ! said the said the fish the fish in

fish in the in the pot the pot . pot . they

. they should they should not should not fly

not fly kites fly kites in kites in a in a house

a house ! house ! they ! they should

they should not should not . not . Oh . Oh ,

Oh , the , the things the things they

things they will they will bump will bump !

bump ! Oh ! Oh , Oh , the , the things

the things they things they will they will hit

will hit ! hit ! Oh ! Oh , Oh , I , I do I do not

do not like not like it like it ! it ! not

! not one not one little one little bit little bit !

bit ! then ! then Sally then Sally and

Sally and I and I saw I saw them

saw them run them run down run down the

down the hall the hall . hall . we . we saw

we saw those saw those two those two things

two things bump things bump their

bump their kites their kites on kites on the

on the wall the wall ! wall ! bump ! bump !

bump ! Thump ! Thump ! Thump ! Thump

! Thump ! Thump ! bump ! bump !

bump ! down ! down the down the wall

the wall in wall in the in the hall the hall .

hall . Thing . Thing two Thing two and

two and Thing and Thing one Thing one !

one ! they ! they ran they ran up ran up !

up ! they ! they ran they ran down ran down !

down ! on ! on the on the string the string of

string of one of one kite one kite we

kite we saw we saw mother's saw mother's new
mother's new gown new gown ! gown ! her
! her gown her gown with gown with the
with the dots the dots that dots that are
that are pink are pink , pink , white
, white and white and red and red . red . then
. then we then we saw we saw one
saw one kite one kite bump kite bump on
bump on the on the head the head of
head of her of her bed her bed ! bed ! then

! then those then those things those things ran
things ran about ran about with about with big
with big bumps big bumps , bumps , jumps
, jumps and jumps and kicks and kicks and
kicks and with and with hops with hops and
hops and big and big thumps big thumps and
thumps and all and all kinds all kinds of
kinds of bad of bad tricks bad tricks .
tricks . and . and I and I said I said , said , I
, I do I do not do not like not like the

like the way the way that way that they

that they play they play ! play ! if ! if mother

if mother could mother could see

could see this see this , this , Oh , Oh ,

Oh , what , what would what would she

would she say she say ! say ! then ! then our

then our fish our fish said fish said ,

said , look , look ! look ! look ! look !

look ! and ! and our and our fish

our fish shook fish shook with shook with fear

with fear . fear . your . your mother

your mother is mother is on is on her

on her way her way home way home !

home ! do ! do you do you hear you hear ?

hear ? Oh ? Oh , Oh , what , what will

what will she will she do she do to do to us

to us ? us ? what ? what will what will she

will she say she say ? say ? Oh ? Oh ,

Oh , she , she will she will not will not like

not like it like it to it to find to find us

find us this us this way this way ! way ! so

! so do so do something do something !

something ! fast ! fast ! fast ! said ! said the

said the fish the fish . fish . do . do you

do you hear you hear ! hear ! I ! I saw

I saw her saw her . her . your . your mother

your mother ! mother ! your ! your mother

your mother is mother is near is near !

near ! so ! so , so , as , as fast as fast as

fast as you as you can you can , can , Think

, Think of Think of something of something to
something to do to do ! do ! you ! you will
you will have will have to have to get
to get rid get rid of rid of Thing of Thing one
Thing one and one and Thing and Thing two
Thing two ! two ! so ! so , so , as , as fast
as fast as fast as I as I could I could , could , I
, I went I went after went after my
after my net my net . net . and . and I
and I said I said , said , with , with my

with my net my net I net I can I can get

can get them get them I them I bet I bet .

bet . I . I bet I bet , bet , with , with my

with my net my net , net , I , I can I can get

can get those get those things those things yet

things yet ! yet ! then ! then I then I let

I let down let down my down my net my net .

net . it . it came it came down came down with

down with a with a PLOP a PLOP !

PLOP ! and ! and I and I had I had them

had them ! them ! at ! at last at last !

last ! those ! those two those two things

two things had things had to had to stop

to stop . stop . then . then I then I said

I said to said to the to the cat the cat ,

cat , now , now you now you do you do as

do as I as I say I say . say . you . you pack

you pack up pack up those up those things

those things and things and you and you take

you take them take them away them away !

away! Oh ! Oh dear Oh dear ! dear! said

! said the said the cat the cat . cat . you

. you did you did not did not like not like our

like our game our game . game Oh

. Oh dear Oh dear . dear . what . what a

what a shame a shame ! shame ! what ! what a

what a shame a shame ! shame ! what ! what a

what a shame a shame ! shame ! then

! then he then he shut he shut up shut up the

up the things the things in things in the

in the box the box with box with the
with the hook the hook . hook . and . and the
and the cat the cat went cat went away
went away with away with a with a sad
a sad kind sad kind of kind of look of look .
look . that . that is that is good is good ,
good , said , said the said the fish the fish .
fish . he . he has he has gone has gone away
gone away . away . Yes . Yes . Yes . But
. But your But your mother your mother will

mother will come will come . come . she

. she will she will find will find this

find this big this big mess big mess !

mess ! and ! and this and this mess

this mess is mess is so is so big so big and

big and so and so deep so deep and

deep and so and so tall so tall , tall , we

, we can we can not can not pick not pick it

pick it up it up . up . there . there is

there is no is no way no way at way at all

at all ! all ! and ! and then and then !

then ! Who ! Who was Who was back

was back in back in the in the house

the house ? house ? Why ? Why , Why , the

, the cat the cat ! cat ! have ! have no

have no fear no fear of fear of this

of this mess this mess , mess , said , said the

said the cat the cat in cat in the in the hat

the hat . hat . I . I always I always pick

always pick up pick up all up all my

all my playthings my playthings and

playthings and so and so . so I

. I will I will show will show you

show you another you another good

another good trick good trick that trick that I

that I know I know ! know ! then ! then we

then we saw we saw him saw him pick

him pick up pick up all up all the

all the things the things that things that were

that were down were down . down . he

. he picked he picked up picked up the

up the cake the cake , cake , and , and the

and the rake the rake , rake , and , and the

and the gown the gown , gown , and , and the

and the milk the milk , milk , and , and the

and the strings the strings , strings , and

, and the and the books the books ,

books , and , and the and the dish the dish ,

dish , and , and the and the fan the fan ,

fan , and , and the and the cup the cup ,

cup , and , and the and the ship the ship ,

ship , and , and the and the fish the fish .

fish . and . and he and he put he put them

put them away them away . away . then

. then he then he said he said , said , that

, that is that is that is that . that . and

. and then and then he then he was

he was gone was gone with gone with a

with a tip a tip of tip of his of his hat

his hat . hat . then . then our then our mother

our mother came mother came in came in and
in and she and she said she said to said to us
to us two us two , two , did , did you
did you have you have any have any fun
any fun ? fun ? tell ? tell me tell me .
me . what . what did what did you did you do
you do ? do ? and ? and Sally and Sally and
Sally and I and I did I did not did not know
not know what know what to what to say
to say . say . should . should we should we tell

we tell her tell her the her the things

the things that things that went that went on

went on there on there that there that day

that day ? day ? should ? should we

should we tell we tell her tell her about

her about it about it ? it ? now ? now ,

now , what , what should what should we

should we do we do ? do ? Well ? Well .

Well what . what would

what would you would you do you do if

do if your if your mother your mother asked

mother asked you asked you ?